

HL-LHC v1.9. E = 7.0 TeV. CC = $-190.0 \mu\text{rad}$. $N_b \simeq 2.2 \times 10^{11}$ ppb,

$\beta_{x,1}^* = 0.5 \text{ m}$, $\beta_{y,1}^* = 0.5 \text{ m}$, polarity $\text{IP}_{2/8} = 1/1$

$\Phi/2_{1(\text{H})} = 250 \mu\text{rad}$, $\Phi/2_{5(\text{V})} = 250 \mu\text{rad}$, $\Phi/2_{2,\text{V}} = -170 \mu\text{rad}$, $\Phi/2_{8,\text{V}} = -200 \mu\text{rad}$

$\sigma_z = 7.6 \text{ cm}$, $\varepsilon_n = 2.5 \mu\text{m}$, $Q' = 15$, $I_{\text{MO}} = -235 \text{ A}$, $C^- = 0$

8b4e_1972b_1960_1178_1886_224bpi_12inj. Bunch 1963.

